

Custom Wake-Up Response

Custom Wake-Up Response

1. Course Content
2. Preparing Audio Files
3. Loading Audio Files

1. Course Content

- Add audio files to the multi_brains program's audio library to customize the voice response after wake-up.

[!NOTE]

- This section of the tutorial is only for users who need to customize personalized voice responses and does not affect normal use.
- If you do not need to customize personalized responses, you can skip this section.

2. Preparing Audio Files

- The **audio materials for voice replies can be downloaded and prepared independently.**
- Alternatively, you can generate speech using the system's built-in `generate_voice` command. The speech generation uses the speech synthesis model from the Alibaba Cloud Model Studio Platform, so you need to configure the `ALIYUN_API_KEY` first, as described in the [02 - Configuring API-KEY] section of this chapter.
- Pre-configure the Alibaba Cloud Model Studio Platform API-KEY



```
*~/M3Pro_ws/multi_brains_file/multi_brains_setting.yaml - Mousepad
File Edit Search View Document Help
#####
#common setting
#通用设置项
#####
ALIYUN_API_KEY: 'sk-82f1a8535fdd43bbaca913e9xxxxxxx'      #阿里云API_KEY Alibaba Cloud API_KEY
LANGUAGE: 'zh'      #Language zh: Chinese and en: English
DEBUG_MODE: False  #Debugging mode
#####
# dify setting
# dify配置选项
#####
DIFY_BASE_URL: "http://localhost/v1"
DIFY_API_KEY: "app-mhawRyoHteauIho7wvXhIhwR"      #dify applies API_KEY
#####
#ASR function setting
#语音识别功能设置
#####
USE_ONLINE_ASR : True      # Whether to use online ASR
ASR_SUPPLIER : 'aliyun'    #ASR Supplier (only effective when using online ASR): Chinese mainland: aliyun International: :
ONLINE_ASR_MODEL : 'paraformer-realtime-v2'
ASR_THRESHOLD : 3      #ASR识别结果阈值, 单位: 字符ASR recognition result threshold. Unit: characters
WAKEUP_THRESHOLD : 5.0  #唤醒时间阈值, 防止WAKEUP_THRESHOLD时间内被多次唤醒, 单位: 秒 Wake-up time threshold, preventing mult
VAD_MODE: 1      #vad灵敏度 VAD sensitivity
MAX_SILENCE_FRAMES: 90  #尾音时长检测, 单位: 帧数 Tail sound duration detection, unit: frames
#####
#TTS function setting
#语音合成功能设置
#####
USE_ONLINE_TTS : True      #Whether to use online TTS
TTS_SUPPLIER : 'aliyun'    # Supplier (only effective when using online TTS): Chinese mainland:
```

- Run the command in the terminal:

```
generate_vioce --text Hello --language_type English
```

- `--text` is the startup parameter; replace "Hello" with the text you want to synthesize into speech.

```
[System Information]
-----
IP_Address_1: 192.168.8.88
IP_Address_2: 192.168.2.47
MACHINE: OrinNx | ROS_DISTRO: humble | ROS_DOMAIN_ID: 30
ROBOT_TYPE: ROSMASTER-M3Pro | CAMERA_TYPE: dabai_dcw2 | RADAR: Tmini-plus*2
-----
jetson@yahboom:~$ generate_vioce --text 你好
The audio file has been saved to: /home/jetson/generate_voices/audio.wav
jetson@yahboom:~$
```

- The audio file will be automatically saved in the `~/generate_voices/` directory.

[!NOTE]

Other optional startup parameters are as follows:

- `--voice`: Speaker, default `Cherry`
- `--language_type`: Language, specify the language of the synthesized audio. Available options include `Chinese`, `English`, `German`, `Italian`, `Portuguese`, `Spanish`, `Japanese`, `Korean`, `French`, and `Russian`.
- `--save_path`: Audio file save path, default `~/generate_voices/`
- `--config_file`: Configuration file path, default `~/M3Pro_ws/multi_brains_file/multi_brains_setting.yaml`
- `--text`: Text to be synthesized audio, default is empty
- `--model` The speech synthesis model defaults to `qwen3-tts-flash`.

For available speakers and speech synthesis models, please refer to the dynamic notifications on the Bailian official website: https://bailian.console.aliyun.com/?spm=5176.29619931.J_SEsSjsNv72y.RuRFS2VknO.2.74cd10d73l2Pw5&tab=doc#/doc/?type=model&url=2879134

- Reference model:

Model availability

We recommend Qwen3-TTS-Flash.

Qwen3-TTS-Flash offers 49 voices and supports multiple languages and dialects.

Qwen-TTS offers up to 7 voices and supports only Chinese and English.

International (Singapore)China (Beijing)	
Model	Version
qwen3-tts-flash Capabilities are identical to qwen3-tts-flash-2025-09-18.	Stable
qwen3-tts-flash-2025-11-27	Snapshot
qwen3-tts-flash-2025-09-18	Snapshot

- Reference Tone

Qwen3-TTS-Flash

Supported voices vary by models. Set the `voice` request parameter

View supported voices of qwen3-tts-flash-2025-11-27

Name	<code>voice</code> parameter
Cherry	Cherry
Serena	Serena
Serena	Ethan
Chelsie	Chelsie
Momo	Momo
Vivian	Vivian
Moon	Moon

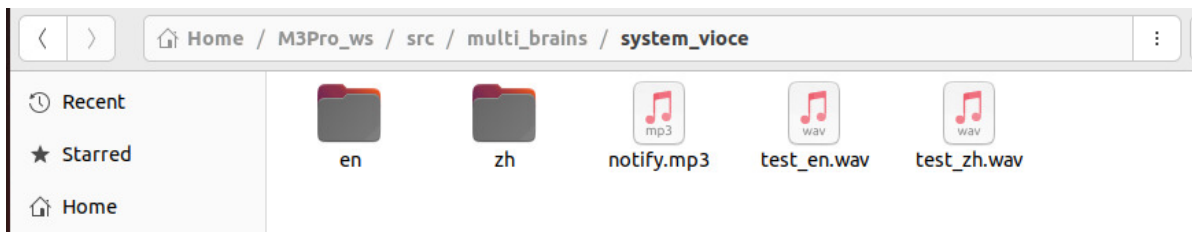
- Supported Languages

Chinese, English, Spanish, Russian, Italian, French, Korean, Japanese, German, Portuguese

3. Loading Audio Files

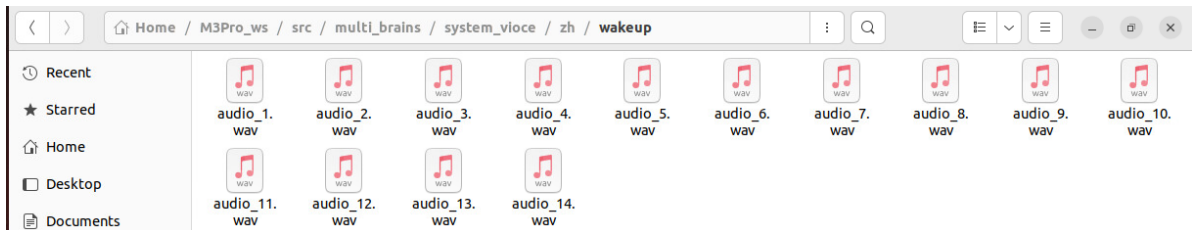
- multi_brains system audio path:

```
~/M3Pro_ws/src/multi_brains/system_vioce
```



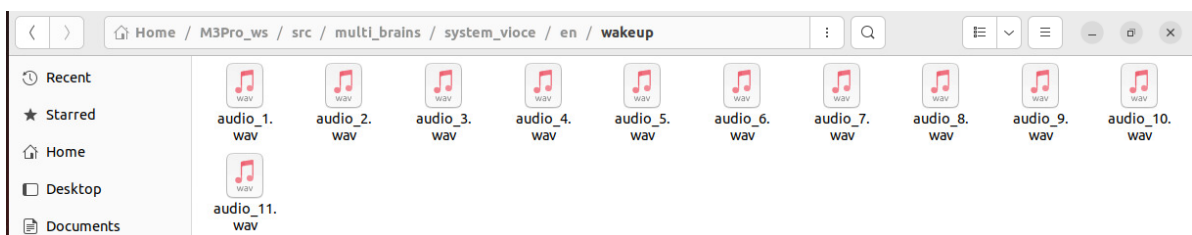
- Where:
- **zh** is the Chinese response voice, suitable for domestic users. Place the prepared audio files in the directory:

```
~/M3Pro_ws/src/multi_brains/system_vioce/zh/wakeup
```



- **en** is the English response voice, suitable for international users. Place the prepared audio files in the directory:

```
~/M3Pro_ws/src/multi_brains/system_vioce/zh/wakeup
```



- When the multi_brains program is started, it will automatically load the audio files in the corresponding directory and randomly play personalized response voices when the user wakes the system.